

Title: AI Copilot for Code Review Using Large Language Models

Abstract

Software code review is an essential activity in modern software development to improve code quality, detect defects, identify security vulnerabilities, and ensure maintainability before deployment. Although recent Large Language Model (LLM)-based code review tools have significantly automated the review process, they often generate generic suggestions without fully understanding repository-level context, providing clear explanations, or effectively integrating traditional static code analysis. These limitations reduce developer trust and may lead to incomplete or inaccurate review recommendations.

This project proposes an **Explainable Repository-Aware AI Copilot for Code Review** that integrates **Code Large Language Models (CodeLlama)**, **Prompt Engineering**, **Static Code Analysis**, and **Retrieval-Augmented Generation (RAG)** to deliver intelligent, context-aware, and explainable code review recommendations. The proposed framework first analyzes the uploaded Java project using static analysis tools such as **SpotBugs** and **PMD** to identify potential code defects and quality issues. Simultaneously, repository-level contextual information is retrieved using a RAG pipeline built with **CodeBERT embeddings** and **FAISS**, enabling the LLM to understand relationships between source files, project structure, and coding standards. The retrieved repository context, together with the static analysis results and developer code, is combined through prompt engineering and processed by **CodeLlama** to generate detailed review comments, suggested fixes, and developer-oriented explanations.

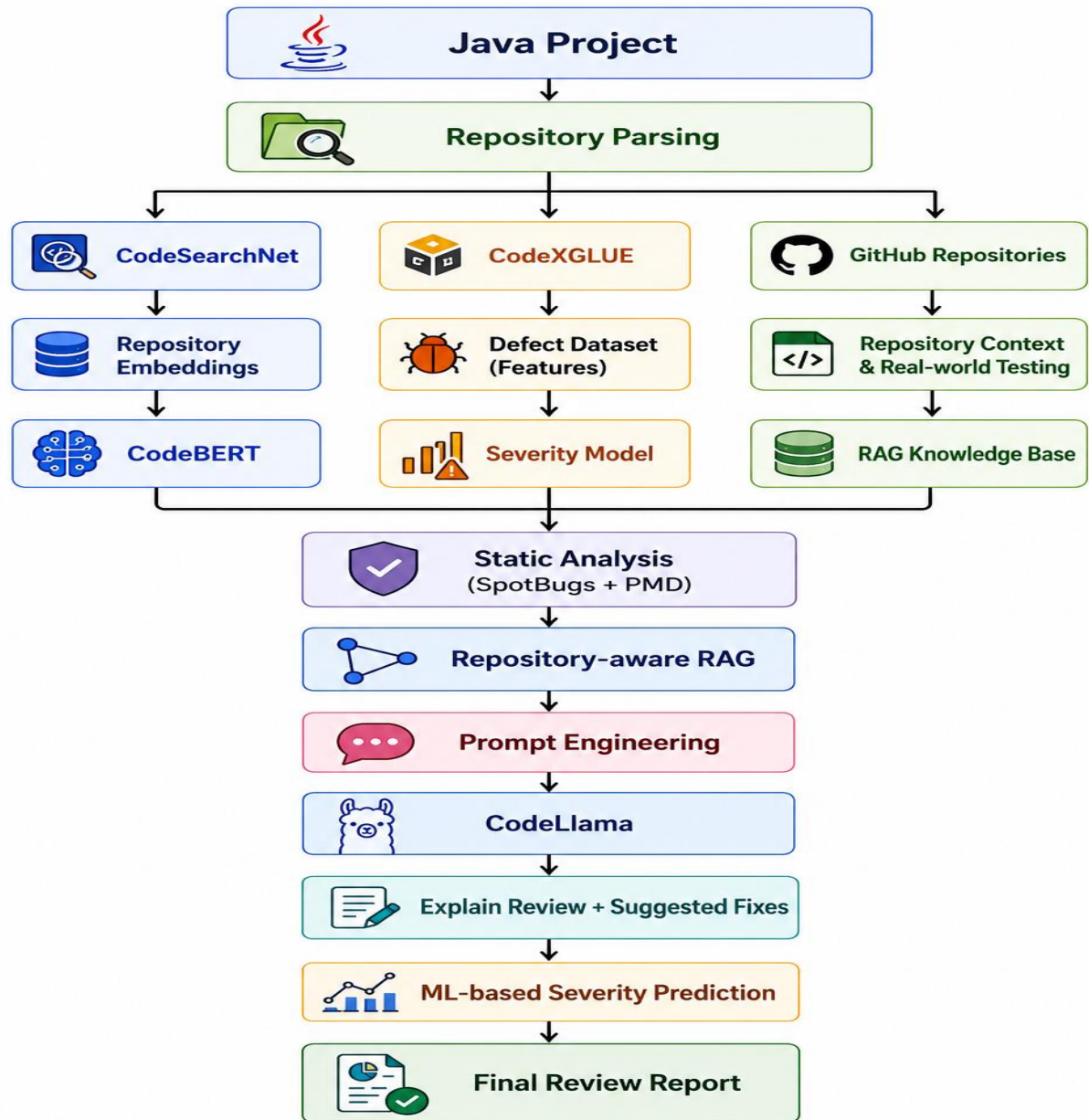
To further enhance review prioritization, the proposed system incorporates a **Machine Learning-based Severity Prediction Model** that classifies detected issues into **Critical, High, Medium, and Low** severity levels using features extracted from static analysis results and repository metadata. The framework also provides developer learning recommendations by retrieving relevant coding standards and best practices to help developers understand and resolve identified issues effectively.

The proposed system utilizes **CodeXGLUE** for defect-related feature extraction and evaluation, **CodeSearchNet** for repository-aware retrieval and embedding generation, and **GitHub Public Repositories** for repository context and real-world evaluation. The expected outcome is an intelligent AI-assisted code review framework capable of generating explainable, repository-aware, and severity-prioritized review recommendations, thereby improving software quality, developer productivity, and the effectiveness of automated code review systems.

Datasets

dataset	links
CodeXGLUE	https://github.com/microsoft/codexglue?utm_source=chatgpt.com
CodeSearchNet	https://github.com/github/CodeSearchNet?utm_source=chatgpt.com
GitHub Public Repositories	https://github.com?utm_source=chatgpt.com

Approach



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